The documentation and process conversion measures necessary to comply with this revision shall be completed by 27 October 2014.

INCH-POUND

MIL-PRF-19500/751A
12 September 2014
SUPERSEDING
MIL-PRF-19500/751
27 October 2008

PERFORMANCE SPECIFICATION SHEET

SEMICONDUCTOR DEVICE, FIELD EFFECT TRANSISTOR,
P-CHANNEL, SILICON, TYPE 2N7508U3,
JANTX, JANTXV, AND JANS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-19500.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for a P-channel, enhancement-mode, MOSFET, power transistor. Three levels of product assurance are provided for each device type as specified in MIL-PRF-19500.

1.2 Physical dimensions. See figure 1, SMD-0.5, TO-276AA (U3).

1.3 Maximum ratings. (Unless otherwise specified, $T_A$ = +25°C).

| Type | $P_T$ (1) $T_C$ = +25°C | $P_T$ $T_A$ = +25°C | $R_{\theta JC}$ (2) | $V_{DS}$ | $V_{DG}$ | $V_{GS}$ | $I_{D1}$ (3) (4) $T_C$ = +25°C | $I_{D2}$ $T_C$ = +100°C | $I_S$ | $I_{DM}$ (5) | $T_J$ and $T_{STG}$ |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | W | W | °C/W | V dc | V dc | V dc | A dc | A dc | A dc | A(pk) | °C |
| 2N7508U3 | 75 | 1.56 | 1.67 | -150 | -150 | ±20 | -11 | -7.2 | -11 | -44 | -55 to +150 |

(1) Derate linearly 0.6 W/°C for $T_C$ > +25°C;
(2) See figure 2, thermal impedance curves.
(3) The following formula derives the maximum theoretical $I_D$ limit. $I_D$ is also limited to 22 A by package, internal construction wires and package pin size.

$$I_D = \sqrt{\frac{T_{JM} - T_C}{(R_{\theta JC}) \times (R_{DS}(on) \text{ at } T_{JM})}}$$

(4) See figure 3, maximum drain current graph.
(5) $I_{DM}$ = 4 X $I_{D1}$ as calculated in note (3).

Comments, suggestions, or questions on this document should be addressed to DLA Land and Maritime, ATTN: VAC, P.O. Box 3990, Columbus, OH 43218-3990, or emailed to Semiconductor@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil .



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T05:30Z
Check the source to verify that this is the current version before use.

1.4 Maximum ratings. Unless otherwise specified, $T_C$ = +25°C.

| Type | Min $V_{(BR)DSS}$ $V_{GS}$ = 0 $I_D$ = -0.25 mA dc | $V_{GS(TH)}$ $V_{DS} \geq V_{GS}$ $I_D$ = -0.25 mA dc | | Max $I_{DSS1}$ $V_{GS}$ = 0 $V_{DS}$ = 100 percent of rated $V_{DS}$ | Max $r_{DS(ON)}$ (1) $V_{GS}$ = -10 V dc | | $E_{AS}$ at $I_{AS}$ | $I_{AS}$ |
|---|---|---|---|---|---|---|---|---|
| | | | | | $T_J$ = +25°C at $I_{D2}$ | $T_J$ = +150°C at $I_{D2}$ | | |
| | V dc | V dc | | µA dc | ohm | ohm | mJ | A |
| | | Min | Max | | | | | |
| 2N7508U3 | -150 | -2.0 | -4.0 | -25 | 0.290 | 0.585 | 130 | -7.2 |

(1) Pulsed (see 4.5.1).

2. APPLICABLE DOCUMENTS

* 2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

DEPARTMENT OF DEFENSE STANDARDS

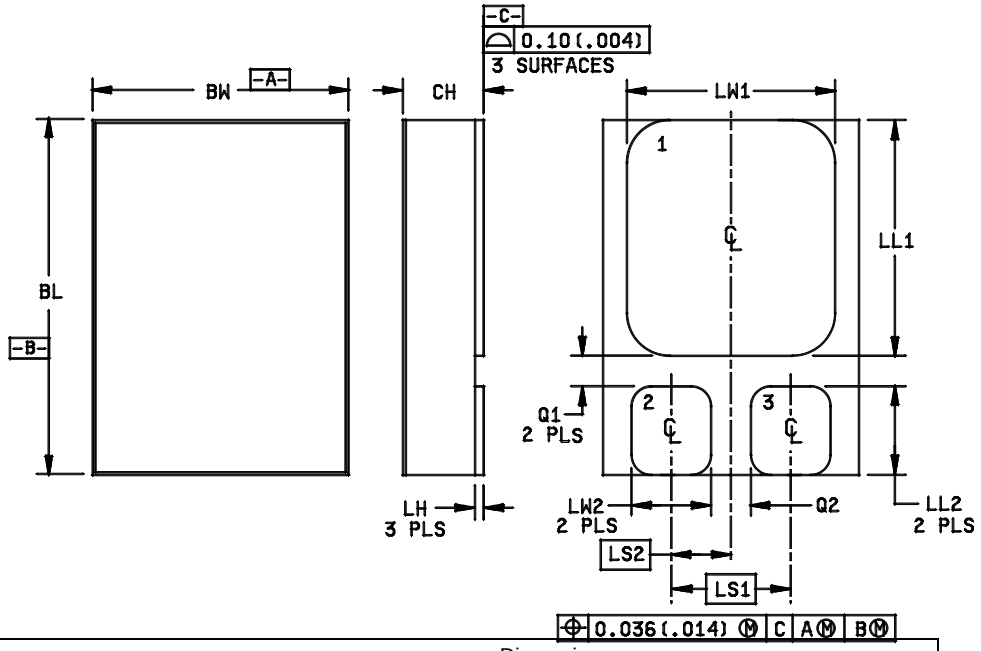
MIL-STD-750 - Test Methods for Semiconductor Devices.

* (Copies of these documents are available online http://quicksearch.dla.mil

2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T05:30Z
Check the source to verify that this is the current version before use.




| Dimensions | | | | |
|---|---|---|---|---|
| Ltr. | Inches | | Millimeters | |
| | Min | Max | Min | Max |
| BL | .395 | .405 | 10.04 | 10.28 |
| BW | .291 | .301 | 7.40 | 7.64 |
| CH | .1085 | .123 | 2.76 | 3.12 |
| LH | .010 | .020 | 0.25 | 0.51 |
| LW1 | .281 | .291 | 7.14 | 7.39 |
| LW2 | .090 | .100 | 2.29 | 2.54 |
| LL1 | .220 | .230 | 5.59 | 5.84 |
| LL2 | .115 | .125 | 2.93 | 3.17 |
| LS1 | .150 BSC | | 3.81 BSC | |
| LS2 | .075 BSC | | 1.91 BSC | |
| Q1 | .030 | | 0.762 | |
| Q2 | .030 | | 0.762 | |
| TERM 1 | Drain | | | |
| TERM 2 | Gate | | | |
| TERM 3 | Source | | | |

NOTES:
1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. In accordance with ASME Y14.5M, diameters are equivalent to ϕx symbology.
4. Terminal 1 - Drain, Terminal 2 - Gate, Terminal 3 - Source.

FIGURE 1. Physical dimensions for SMD-0.5 TO-276AA (2N7508U3).



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T05:30Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

3.1 General. The individual item requirements shall be as specified in MIL-PRF-19500 and as modified herein.

* 3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturer's list before contract award (see 4.2 and 6.3).

3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500 and as follows.

$I_{AS}$ - Rated avalanche current, nonrepetitive.
nC - nano coulomb.

* 3.4 Interface and physical dimensions. The interface and physical dimensions shall be as specified in MIL-PRF-19500 and on figure 1, SMD-0.5 (TO-276AA) herein.

3.4.1 Lead finish. Lead finish shall be solderable in accordance with MIL-PRF-19500, MIL-STD-750, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.5 Marking. Marking shall be in accordance with MIL-PRF-19500.

3.6 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4, and table I.

3.7 Electrical test requirements. The electrical test requirements shall be as specified in table I.

3.8 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4 and table I).

* 4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

4.2.1 Group E qualification. Group E inspection shall be performed for qualification or re-qualification only. In case qualification was awarded to a prior revision of the specification sheet that did not request the performance of table II tests, the tests specified in table II herein that were not performed in the prior revision shall be performed by the first inspection lot of this revision to maintain qualification.



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T05:30Z
Check the source to verify that this is the current version before use.

* 4.3 Screening (JANS, JANTX and JANTXV levels only). Screening shall be in accordance with table E-IV of MIL-PRF-19500 and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

| Screen (see table E-IV of MIL-PRF-19500) (1) (2) | Measurement | |
|---|---|---|
| | JANS level | JANTX and JANTXV levels |
| (3) | Gate stress test (see 4.3.1) | Gate stress test (see 4.3.1) |
| (3) (4) | Method 3470 of MIL-STD-750 (see 4.3.2) optional | Method 3470 of MIL-STD-750 (see 4.3.2) optional |
| (3) 3c | Method 3161 of MIL-STD-750 (see 4.3.3) | Method 3161 of MIL-STD-750 (see 4.3.3) |
| 9 | $I_{GSSF1}$, $I_{GSSR1}$, $I_{DSS1}$, subgroup 2 of table I herein | Not applicable |
| 10 | Method 1042 of MIL-STD-750, test condition B | Method 1042 of MIL-STD-750, test condition B |
| 11 | Subgroup 2 of table I herein;<br>$I_{GSSF1}$, $I_{GSSR1}$, $I_{DSS1}$, $r_{DS(on)1}$,$V_{GS(TH)1}$,<br>$\Delta I_{GSSF1}$ = ±20 nA dc or ±100 percent of initial value, whichever is greater.<br>$\Delta I_{GSSR1}$ = ±20 nA dc or ±100 percent of initial value, whichever is greater.<br>$\Delta I_{DSS1}$ = ±25 μA dc or ±100 percent of initial value, whichever is greater. | Subgroup 2 of table I herein,<br>$I_{GSSF1}$, $I_{GSSR1}$, $I_{DSS1}$, $r_{DS(on)1}$,$V_{GS(TH)1}$ |
| 12 | Method 1042 of MIL-STD-750, test condition A | Method 1042 of MIL-STD-750, test condition A |
| 13 | Subgroups 2 and 3 of table I herein.<br>$\Delta I_{GSSF1}$ = ±20 nA dc or ±100 percent of initial value, whichever is greater.<br>$\Delta I_{GSSR1}$ = ±20 nA dc or ±100 percent of initial value, whichever is greater.<br>$\Delta I_{DSS1}$ = ±25 μA dc or ±100 percent of initial value, whichever is greater.<br>$\Delta r_{DS(on)1}$ = ±20 percent of initial value.<br>$\Delta V_{GS(TH)1}$ = ±20 percent of initial value. | Subgroup 2 of table I herein.<br>$\Delta I_{GSSF1}$ = ±20 nA dc or ±100 percent of initial value, whichever is greater.<br>$\Delta I_{GSSR1}$ = ±20 nA dc or ±100 percent of initial value, whichever is greater.<br>$\Delta I_{DSS1}$ = ±25 μA dc or ±100 percent of initial value, whichever is greater.<br>$\Delta r_{DS(on)1}$ = ±20 percent of initial value.<br>$\Delta V_{GS(TH)1}$ = ±20 percent of initial value. |
| * 17 | For U3 packages: Method 1081 of MIL-STD-750 (see 4.3.4), Endpoints: Subgroup 2 of table I herein. | For U3 packages: Method 1081 of MIL-STD-750 (see 4.3.4), Endpoints: Subgroup 2 of table I herein. |

(1) At the end of the test program, $I_{GSSF1}$ $I_{GSSR1}$, and $I_{DSS1}$ are measured.
* (2) An out-of-family program to characterize $I_{GSSF1}$, $I_{GSSR1}$, $I_{DSS1}$, $V_{GS(th)1}$ and $_{RDS(ON)1}$ shall be invoked.
(3) Shall be performed anytime after temperature cycling, screen 3a; TX and TXV devices do not need to be repeated in screening requirements.
(4) Method 3470 is optional if performed as a sample in group A, subgroup 5.



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T05:30Z
Check the source to verify that this is the current version before use.

4.3.1 Gate stress test. Apply $V_{GS}$ = 24 V minimum for t = 250 µs minimum.

4.3.2 Single pulsed unclamped inductive switching.

a. Peak current, $I_D$ ..... $I_{AS(max)}$.

b. Peak gate voltage, $V_{GS}$ ..... -10 V, up to maximum rated $V_{GS}$

c. Gate to source resistor, $R_{GS}$ ..... $25 \leq R_g \leq 200\Omega$.

d. Initial case temperature ..... +25°C, +10°C, -5°C.

e. Inductance, L ..... $\left[\frac{2E_{AS}}{(I_{D1})^2}\right]\frac{[(V_{BR}-V_{DD})]}{V_{BR}}$ *mH minimum.*

f. Number of pulses to be applied ..... 1 pulse.

g. Supply voltage ($V_{DD}$) ..... ……-50 V.

4.3.3 Thermal impedance. The thermal impedance measurements shall be performed in accordance with method 3161 of MIL-STD-750 using the guidelines in that method for determining $I_M$, $I_H$, $t_H$, $t_{SW}$, (and $V_H$ where appropriate). Measurement delay time ($t_{MD}$) = 70 µs max. See table II, group E, subgroup 4 herein.

* 4.3.4 Dielectric withstanding voltage.

a. Magnitude of test voltage.....600 V dc.

b. Duration of application of test voltage.....15 seconds (min).

c. Points of application of test voltage .....All terminals to case.

d. Method of connection.....Mechanical.

e. Kilovolt-ampere rating of high voltage source .....1,200 V/1.0 mA (min).

f. Maximum leakage current.....1.0 mA.

g. Voltage ramp up time.....500 V/second

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500. Alternate flow is allowed for conformance inspection in accordance with MIL-PRF-19500.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-PRF-19500 and table I herein. Electrical measurements (end-points) shall be in accordance with the inspections of table I, subgroup 2 herein.



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T05:30Z
Check the source to verify that this is the current version before use.

* 4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-VIA (JANS) and table E-VIB (JANTX and JANTXV) of MIL-PRF-19500 and as follows.

4.4.2.1 Group B inspection table E-VIA (JANS) of MIL-PRF-19500.

| Subgroup | Method | Conditions |
|---|---|---|
| B3 | 1051 | Test condition G. |
| * B4 | 1042 | Test condition D. The heating cycle shall be 1 minute minimum. No heat sink nor forced air cooling on the device shall be permitted during the "on" cycle. |
| B5 | 1042 | A separate sample may be pulled for each test. Accelerated steady-state reverse bias; test condition A, $V_{DS}$ = rated, $T_A$ = +175°C, t = 120 hours, read and record $V_{BR(DSS)}$ (pre and post) at $I_D$ = -1 mA. Read and record $I_{DSS}$ (pre and post). |
| B5 | 1042 | Accelerated steady-state gate stress; test condition B, $V_{GS}$ = rated, $T_A$ = +175°C, t = 24 hours. |
| B5 | 2037 | Test condition D. |

4.4.2.2 Group B inspection table E-VIB (JANTX and JANTXV) of MIL-PRF-19500.

| Subgroup | Method | Conditions |
|---|---|---|
| B2 | 1051 | Test condition G. |
| * B3 | 1042 | Test condition D. The heating cycle shall be 1 minute minimum. No heat sink nor forced air cooling on the device shall be permitted during the "on" cycle. |

* 4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-VII of MIL-PRF-19500 and as follows.

4.4.3.1 Group C inspection (table E-VII of MIL-PRF-19500).

| Subgroup | Method | Conditions |
|---|---|---|
| C2 | 2036 | Not applicable. |
| C5 | 3161 | $R_{\theta JC}$ = 1.67 °C/W maximum. See 4.5.2. |
| * C6 | 1042 | Test condition D. The heating cycle shall be 1 minute minimum. No heat sink nor forced air cooling on the device shall be permitted during the "on" cycle. |

* 4.4.4 Group E inspection. Group E inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-IX of MIL-PRF-19500 and as specified in table II herein.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

4.5.2 Thermal resistance. The thermal resistance measurements shall be performed in accordance with method 3161 of MIL-STD-750 using the guidelines in that method for determining $I_M$, $I_H$, $t_H$, $t_{SW}$ (and $V_H$ where appropriate). Measurement delay time ($t_{MD}$) = 70 μs max. See table E-IX of MIL-PRF-19500, group E, subgroup 4.



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T05:30Z
Check the source to verify that this is the current version before use.

TABLE I. Group A inspection.

| Inspection 1/ | MIL-STD-750 | | Symbol | Limits | | Unit |
|---|---|---|---|---|---|---|
| | Method | Conditions | | Min | Max | |
| Subgroup 1 | | | | | | |
| Visual and mechanical inspection | 2071 | | | | | |
| Subgroup 2 | | | | | | |
| Thermal impedance 2/ | 3161 | See 4.3.3 | $Z_{\theta JC}$ | | | °C/W |
| Breakdown voltage, drain to source | 3407 | $V_{GS}$ = 0 V dc, $I_D$ = -0.25 mA dc, condition C | $V_{(BR)DSS}$ | -150 | | V dc |
| Gate to source voltage (threshold) | 3403 | $V_{DS} \geq V_{GS}$, $I_D$ = -0.25 mA dc | $V_{GS(th)1}$ | -2.0 | -4.0 | V dc |
| Gate current | 3411 | $V_{GS}$ = -20 V dc, $V_{DS}$ = 0, bias condition C | $I_{GSSF1}$ | | -100 | nA dc |
| Gate current | 3411 | $V_{GS}$ = +20 V dc, $V_{DS}$ = 0, bias condition C | $I_{GSSR1}$ | | +100 | nA dc |
| Drain current | 3413 | $V_{GS}$ = 0 V dc, $V_{DS}$ = 100 percent of rated $V_{DS}$; bias condition C | $I_{DSS1}$ | | -25 | µA dc |
| Static drain to source on-state resistance | 3421 | $V_{GS}$ = 10 V dc, pulsed (see 4.5.1); condition A, $I_D$ = rated $I_{D2}$ (see 1.3) | $r_{DS(on)1}$ | | 0.290 | Ω |
| Forward voltage (source drain diode) | 4011 | Pulsed (see 4.5.1), $I_S$ = rated $I_{D2}$ | $V_{SD}$ | | -1.6 | V |
| Subgroup 3 | | | | | | |
| High temperature operation: | | $T_C = T_J$ = +125°C | | | | |
| Gate to source voltage threshold | 3403 | $V_{DS} \geq V_{GS}$, $I_D$ = -0.25 mA dc | $V_{GS(th)2}$ | -1.0 | | V dc |
| Gate current | 3411 | Bias condition C, $V_{GS}$ = ±20 V dc, $V_{DS}$ = 0 V dc | $I_{GSS2}$ | | ±200 | nA dc |
| Drain current | 3413 | Bias condition C, $V_{GS}$ = 0 V dc, $V_{DS}$ = 80 percent of rated $V_{DS}$ | $I_{DSS2}$ | | -0.25 | mA dc |
| Static drain to source on-state resistance | 3421 | $V_{GS}$ = 10 V dc, pulsed (see 4.5.1); $I_D$ = rated $_{ID}2$ | $r_{DS(on)2}$ | | 0.522 | Ω |
| Low temperature operation: | | $T_C = T_J$ = -55°C | | | | |
| Gate to source voltage threshold | 3403 | $V_{DS} \geq V_{GS}$, $I_D$ = -0.25 mA dc | $V_{GS(th)3}$ | | -5.0 | V dc |

See footnotes at end of table.



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T05:30Z
Check the source to verify that this is the current version before use.

TABLE I. Group A inspection - Continued.

| Inspection 1/ | MIL-STD-750 | | Symbol | Limits | | Unit |
|---|---|---|---|---|---|---|
| | Method | Conditions | | Min | Max | |
| Subgroup 4 | | | | | | |
| Forward transconductance | 3475 | $I_D$ = rated $I_{D2}$, $V_{DD}$ = -25 V (see 4.5.1) | $g_{FS}$ | 2.3 | | S |
| Switching time test | 3472 | $I_D$ = rated $I_{D2}$, $V_{GS}$ = -10 V dc, gate drive impedance = 6.8 Ω $V_{DD}$ = 50% rated $V_{DS}$ | | | | |
| Turn-on delay time | | | $t_{d(on)}$ | | 25 | ns |
| Rise time | | | $t_r$ | | 65 | ns |
| Turn-off delay time | | | $t_{d(off)}$ | | 75 | ns |
| Fall time | | | $t_f$ | | 53 | ns |
| Subgroup 5 | | | | | | |
| Safe operating area test | 3474 | See figure 4; $V_{DS}$ = 80 percent of rated $V_{BR(DSS)}$, $V_{DS} \leq$ -200 V; tp = 10 ms | | | | |
| Electrical measurements | | See table I, subgroup 2 | | | | |
| Subgroup 6 | | | | | | |
| Not applicable | | | | | | |
| Subgroup 7 | | | | | | |
| Gate charge | 3471 | Condition B, $I_D$ = rated $I_{D2}$, $V_{DD}$ = 80 percent rated $V_{DS}$ | Qg(on) | | 66 | nC |
| On-state charge | | | | | | |
| Gate to source charge | | | Qgs | | 13 | nC |
| Gate to drain charge | | | Qgd | | 40 | nC |
| Reverse recovery time | 3473 | di/dt = 100 A/µs, $V_{DD} \leq$ -25 V, $I_D$ = rated $I_{D2}$ | trr | | 240 | ns |

1/ For sampling plan, see MIL-PRF-19500.
2/ This test required for the following end-point measurements only:
Group B, subgroups 2 and 3 (JANTX and JANTXV).
Group B, subgroups 3 and 4 (JANS).
Group C, subgroup 2 and 6.
Group E, subgroup 1.



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T05:30Z
Check the source to verify that this is the current version before use.

TABLE II. Group E inspection (all quality levels) for qualification or re-qualification only.

| Inspection | MIL-STD-750 | | Sample plan |
|---|---|---|---|
| | Method | Conditions | |
| Subgroup 1 | | | 45 devices c = 0 |
| Temperature cycle | 1051 | Condition G, 200 cycles | |
| Hermetic seal<br>Fine leak<br>Gross leak | 1071 | | |
| Electrical measurements | | See table I, subgroup 2 | |
| Subgroup 2 | | | 45 devices c = 0 |
| Steady-state reverse bias | 1042 | Condition A, 1,000 hours | |
| Electrical measurements | | See table I, subgroup 2 | |
| Steady-state gate bias | 1042 | Condition B, 1,000 hours, $V_{GS}$ = 80 percent of rated (see 1.3) | |
| Electrical measurements | | See table I, subgroup 2 | |
| Subgroup 4 | | | sample size N/A |
| Thermal impedance curves | | See MIL-PRF-19500 | |
| Subgroup 10 | | | |
| Commutating diode for safe operating area test procedure for measuring dv/dt during reverse recovery of power MOSFET transistors or insulated gate bipolar transistors | 3476 | Test conditions shall be derived by the manufacturer | 22 devices c = 0 |



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T05:30Z
Check the source to verify that this is the current version before use.

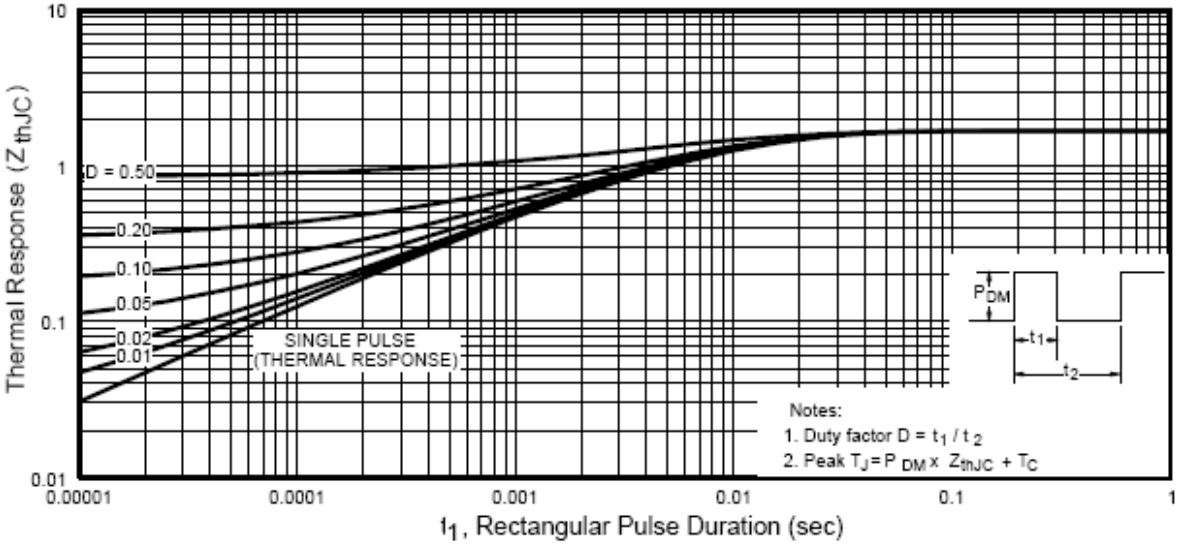



FIGURE 2. Thermal impedance curves.

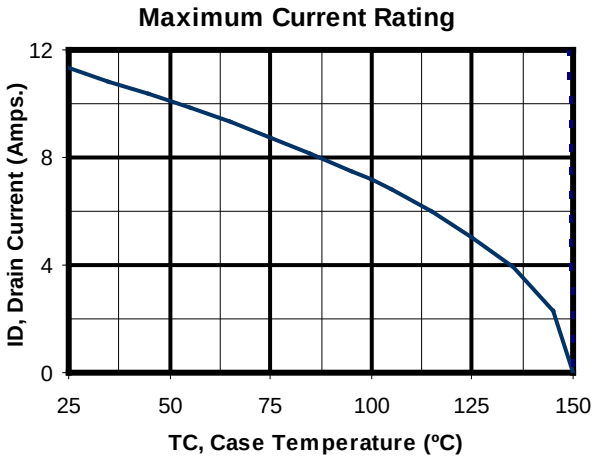



FIGURE 3. Maximum drain current versus case temperature graphs.



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T05:30Z
Check the source to verify that this is the current version before use.

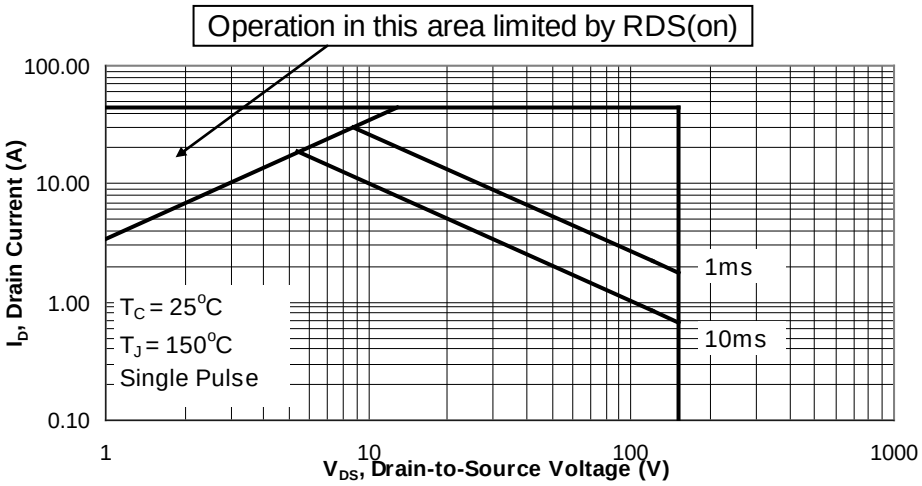



FIGURE 4. Maximum safe operating area.



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T05:30Z
Check the source to verify that this is the current version before use.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory. The notes specified in MIL-PRF-19500 are applicable to this specification.)

6.1 Intended use. Semiconductors conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

* 6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Packaging requirements (see 5.1).

c. Lead finish (see 3.4.1).

d. The complete Part or Identifying Number (PIN), see title and section 1.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List (QML 19500) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from DLA Land and Maritime, ATTN: VQE, P.O. Box 3990, Columbus, OH 43218-3990 or e-mail vqe.chief@dla.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://assist.dla.mil .

6.4 Cross-reference complement list. Parts from this specification may be used to supersede the following commercial Part or Identifying Number (PIN) listed below. The following table shows the generic PIN and its associated military PIN (without JAN prefix).

| Commercial type | Military type |
|---|---|
| IRF5NJ6215 | 2N7508U3 |



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T05:30Z
Check the source to verify that this is the current version before use.

* 6.5 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

Custodians:
Army - CR
Navy - EC
Air Force - 85
NASA - NA
DLA - CC

Preparing activity:
DLA - CC

(Project 5961-2014-113)

* NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T05:30Z
Check the source to verify that this is the current version before use.